

Infrared light emitting LED (IR LED)

Visible light emitting LED

A typical LED has a current rating of 20 mA, life time of 2 lakh working hours, and can tolerate a maximum voltage of around 4 V. There are also coloured LEDs available in the market, and their colour is due to the doping in the semiconducting material used to make them.

All LEDs have two terminals, one positive and the other negative. The negative terminal is of shorter length, but there are some LEDs with a longer negative terminal which may be due to manufacturing defect. Also, once you cut the lengths of a LED to use it in a circuit, the identification by comparing lengths of terminals is not useful. Another way of identifying the positive and negative terminal is to compare the filament. Again, due to

manufacturing defects, some LEDs have a broader positive terminal. So none of the ways mentioned can be relied upon completely. The best way is to use the multimeter. Set the multimeter to continuity checking mode and touch red probe to one terminal of LED and black probe to other terminal of LED. If the LED glows, the terminal in contact with the red probe is the positive terminal and other is the negative one. If the LED doesn't glow, change the polarity. Now, if the LED glows, the terminal now in contact with the red probe is positive and the other negative. However, if the LED still doesn't glow, it means either the LED or the multimeter is faulty.



An LED, with the cathode on the left and anode on the right

LEDs are generally used in large number in circuit. They are used in parallel with a regular connection in a circuit, and this helps a lot during troubleshooting (that is finding the error in the circuit if it doesn't work). An LED glowing in parallel of a connection suggests that the current is flowing through the connection in the direction of the negative terminal from the positive terminal of LED. However, if the LED doesn't glow, it suggests that either the current is flowing in the opposite direction, or the current is not flowing at all in that part of the circuit. A thumb rule to be followed while using LEDs is to use a resistor of around 100 ohm in series with LED, so as to avoid the LED getting burnt in case the voltage across it exceeds 4V or current exceeds 20mA.

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There are two differences between a normal LED and an IR LED. The first